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Agentic AI [Alpha Assign-2]

Question 1)

1.) Bits per frame = 29491200 bits/frame

pixels per frame = 1280 × 720 = 921600 pixels

X = 29491200 ÷ 921600 = 32 bits per pixel.

2.) Bytes per frame = 29491200 ÷ 8 = 3686400 bytes per frame.

3) 70 × 1048576 × 8 ⇒ 587202560 bits per second.

Question 2)

Q =		K =		Q =	
1.5	3	1.1	2.2	0.2	0.3
2	4	1	3	2	0.3
2.2	4.4	1	3	2.2	0.3
2.4	4.4	1.1	3	2.2	0.3

softmax = $(QK^T/\sqrt{d_k})V$ where $d_k = 2$

QK^T =									
1.5	3	1.1	1	1	1.1	⇒	8.25	10.5	10.5	10.65
2	4	2.2	3	3	3		11	14	14	14.2
2.2	4.4						12.1	15.4	15.4	15.62
2.4	4.4						12.32	15.6	15.6	15.84

$QK^T/\sqrt{d_k}$ =				
5.834	7.425	7.425	7.531
7.778	9.899	9.899	10.041
8.556	10.889	10.889	11.045
8.716	11.03	11.031	11.201

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applying softmax matrix becomes :-

0.0614	0.3016	0.3016	0.3353
0.0366	0.3056	0.3056	0.3522
0.0297	0.3062	0.3062	0.3579
0.0301	0.3045	0.3045	0.3609

↳ This multiplied with

0.2	0.3
2	0.3
2.2	0.3
2.2	0.3

we get Attention :-

2.0167	0.3	→ AI
2.0657	0.3	→ Governace
2.0794	0.3	→ Emerging
2.0789	0.3	→ Course

Question 3)

Wind	Humidity	Rain
Weak	High	Yes
Strong	High	Yes
Weak	Low	No
Weak	Low	Yes
Strong	Low	No

where we take

P(Yes) = 0.6

P(No) = 0.4

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1) N.B with laplace smoothing :-

Likelihood	Rain = Yes (n = 3)	= No (n = 2)
P(weak | class)	(2+1)/(3+2) = 0.60	(1+1)/2+2 = 0.5
P(strong | class)	1+1/3+2 = 0.4	1+1/2+2 = 0.5
P(high | class)	2+1/3+2 = 0.6	0+1/2+2 = 0.25
P(low | class)	1+1/3+2 = 0.4	2+1/2+2 = 0.75

worked example for weak, high :-

P(Yes) score = 0.6 × 0.6 × 0.6 = 0.216

P(No) score = 0.4 × 0.5 × 0.25 = 0.05

Normalized = P(Yes) = 0.216/0.266 = 81.2 % P(No) = 100 - 81.2 = 18.8 %

Using this I'm creating table :- (solving on calculator)

Wind	Humidity	Yes Score	No score	P(Yes)	P(No)	Prediction
Weak	High	0.216	0.050	81.2%	18.8%	Rain, hold pond
Strong	High	0.114	0.050	74.2%	25.8%	Rain, hold pond
weak	low	0.144	0.150	49.0%	51.0%	No rain, deliver
Strong	low	0.096	0.150	39.0%	61.0%	No rain, deliver

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a) Entropy & information gain (ID3 Tree)

Entropy $= -0.6\log_2 0.6 - 0.4\log_2 0.4 = 0.971$

For :- Humidity :-

a) High (2Y, 0N)

$H(\text{High}) = 0$

b) low (1Y, 2N)

$H(\text{low}) = -1/3 \log_2(1/3) - 2/3 \log_2(2/3)$

$= 0.918$

$H(S|\text{Humidity}) = 2/5(0) + 3/5(0.918)$

$= 0.551$

I.G (Humidity) $= 0.971 - 0.551 = 0.42$

For :- Wind :-

a) Strong :- (1Y, 1N)

$H(\text{Strong}) = 1$

b) Weak :- (2Y, 1N)

$H(\text{weak}) = -2/3 \log_2(2/3) - 1/3 \log_2(1/3)$

$= 0.918$

$H(S|\text{wind}) = 2/5(1) + 3/5(0.918) = 0.951$

I.G(wind) $= 0.971 - 0.951 = 0.02$

Root = Humidity

Humidity

High → Rain

low → wind

wind

strong → No Rain

Wed → 50%

Wind :-

a) Weak (1Y, 1N)

$H(\text{weak}) = 1$

$2/3(1) + 1/3(0) = 0.667$

$0.918 - 0.667 = 0.252$

b) Strong (0Y, 1N)

$H(\text{Strong}) = 0$

3) logistic regression :-

Encoding: Wind (W=0, S=1), Humidity (L=0, H=1), Rain (Y=1, N=1)

model $= z = b + w_1 \cdot \text{wind} + w_2 \cdot \text{Humidity}$, $p = \sigma(z) = (1/1+e^{-z})$

Gradient of average cross entropy loss : $\partial J/\partial w = (1/n) \Sigma (p_i - y_i) \cdot x_{ij}$

Update rule : $w \leftarrow w - \alpha \cdot \partial J/\partial w$, $\alpha = 1$ & weights starting from 0.

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Iteration 1) (all $p = 0.5$ so residual $p - y$ are $-0.5, -0.5, +0.5, -0.5, +0.5$)

- Bias gradient $= (-0.5 - 0.5 + 0.5 - 0.5 + 0.5)/5 = -0.10$ so $b = 0.10$
- Wind gradient $= (-0.5 + 0.5)/5 = 0$ so $w_1 = 0$
- Humidity gradient $= (-0.5 - 0.5/5) = -0.20$, so $w_2 = 0.2$

Iteration 2) ($z = (0.3, 0.3, 0.1, 0.1, 0.1)$ $p = (0.574, 0.57, 0.525, 0.525, 0.525)$

- Bias gradient $= -0.055$ so $b = 0.155$
- Wind gradient $= +0.020$ so $w_1 = -0.020$
- Humidity gradient $= -0.170$ so $w_2 = 0.370$

weights keep moving after 1st but soon it'll be :-

Case	z	P(Yes)	P(No)	Prediction
Weak, High	8.82	0.999	0.0001	Yes hold
Strong, High	4.63	0.990	0.010	Yes hold
Weak, low	-0.01	0.4975	0.5025	No deliver (Borderline)
Strong, low	-4.21	0.015	0.985	No (deliver).

4) Comparison :-

Case	Naive Bayes	Decision Tree	logistic Regression
Weak, high	Yes (81%)	Yes	Yes (99.99)%
Strong, high	Yes (74%)	Yes	Yes (99.01)%
Weak, low	No (51%)	No (Tie Broken)	No (50.25%)
Strong, low	No (61%)	No	No (98.5%)
Testing Accuracy	80%	80%	80%

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Question 5)

1) Joint Probability distribution

P(infected) = P(healthy) = 0.5

A	B	Probability
Infected	Infected	0.25
Infected	Healthy	0.25
Healthy	Infected	0.25
Healthy	Healthy	0.25

2) Action Table

Time Stamp	Drone Pos	A	B	Percept	Action	Credit
t=0	A	Infected	Infected	[A, infected]	Spray	$0
t=1	A	healthy	Infected	[A, healthy]	move to B	$2
t=2	B	healthy	Infected	[B, infected]	Spray	$2
t=3	B	healthy	Healthy	[B, healthy]	move to A	$4
		Total			=	$8

Question 6)

Paper 1 :- Role of Game theory AI agent

1) Game theory formalizes interaction between agents as games, which lets you model, predict and optimize how rational agents respond strategically. In the paper it served as the reference point: fair game, runs and agents through classic games and compared their behaviour with the game theoretic prediction. This paper framed this as strategic information among multiple agents rather than collaboration as such. The same tools still apply to coordination, since battle of the sexes is a coordination game.

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2) It gives the mathematical foundation for strategic decision-making and a way to interpret how agents reason. It also gives a benchmark. The author argues that applying game theory to multi-LLM interactions uncover biases and emergent behaviour which improves interpretability, fairness, efficiency, legal compliance & fault. They also point to negotiation, pricing, dispute resolution

3) Prisoners Dilemma:- Players gain by defecting or cooperating with mutual cooperation leading to better payoffs.

Battle of sexes → Coordination/commitment game where players prefer outcomes but value coordinating.

The 1st game (P.D) can model cooperation between base stations for spectrum sharing and interference management while B.O.S can model ~~cooperation~~ coordination in network slicing, handovers and edge computing resource allocation.

4) Anthropic : Claude 3.5 sonnet
OpenAI : GPT-4
MistralAI : Mistral Large (123b)
Meta : Llama 3.1 (405b)

Paper 2 :-

1) It is the Tuple of $(S, A, P, R, \Omega, O, \gamma)$

S : hidden states — $P(s'|s,a)$: transition probability
A : Action — $R(s,a)$: reward
Ω : observation s ~~with~~ — $O(o|s',a)$: probability of seeing O after taken a leads to
$\gamma \in [0,1)$; discount factor.

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2) Hard Parameter Sharing :-

- One shared network learns common features for multiple related tasks.
- Can be used in Multi-Agent reinforcement learning so agents can learn from each others ~~reference~~ experiences.
- Situations where resource limitation is required.

Soft parameter sharing :-

- Each tasks maintains its models and parameters
- When agents require independence, useful
- & totally when tasks having different input/output requirements.

3) $L_\theta(\theta_e) = E_{(o,a,r,o') \sim D}\left[\sum_i \| f_o(z_{a_i}, o_i, a_{-i}) - o_i' \|^2 + \lambda_e \sum_i (f_r(z_{a_i}, o_i, a_{-i}) - r_i\right.$

where

- $L_e(\theta_e)$: loss of the encoder_decoder model, with parameters θ_e
- E over $(o,a,r,o') \sim D$: average over transitions sampled from the shared replay buffer D
- $\sum_i$: sum over all agents
- z^i : latent code for agent i encoder depends only on the agents identity, so the code carries agent specific information such as reward function.
- $f_o(z_i, o_{-i}, a_{-i})$: the observation decoder predicts agents i's next information from its latent information taken by observing agents current action.
- f_r : reward decoder predicts r_i and its squared error is weighted by λ_e.
- λ_e : scaling factor that balances information observation and reward prediction

4) The figure shows 2 rewards which are as follows:

i) Exploration Reward :- agents are paid for reaching states, the forward model predicts poorly, meaning unfamiliar states. Its weights decreases during training, so that the environment is explored

ii) Type identification reward :- agents are awarded when agents of the same type get similar representations and penalized when different types get similar ones.

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Question 7)

i) $A = (6, -1, \sqrt{a})$ $\quad (6)(4) + (-1)(x) + (3)(-2) = 0$

$B = (4, x, -2)$ $\quad x = 18$ //

$A \cdot B = 0$

ii) Dots products between vectors = scalar

Hadamard product = vector.

iii) $u = (1, 2)$ $\quad \|u - proj\| = \sqrt{1.5^2 + 1.5^2} = 3\sqrt{2}/2$

$v = (2, 2)$ $\quad u.v = 6$

$u.v = \|v\| \cdot \|u - proj\|$ $\quad (1,2) \cdot (2,2) = 2 + 4 = 6$

Question 8)

given: Agent A : $0.2t + t^2$ $\quad$ Agent C : $t^5 + t^6 + \sin t$

Agent B : $\sin t + \cos t$ $\quad$ Agent D : $t^5 + t^8 + \sin t + \cos t$

$S(t)$ = sum of all 4 agents (t)

$t = 1$:-

$A = (0.2)(1) + (1)^2 = 1.2$ $\quad B = \sin(1) + \cos(1) = 1.017$

$C = (1)^5 + (1)^6 + \sin 1 = 2.017$ $\quad D = (1)^5 + (1)^8 + \sin(1) + \cos(1) = 3.017$

$t = 2$:-

$A = 0.2(2) + (2)^2 = 4.4$ $\quad B = 1.034$ $\quad C = 96.034$ $\quad D = 289.03$

$t = 3$:-

$A = 9.6$ $\quad B = 1.051$ $\quad C = 972.052$ $\quad D = 6805.051$

$t = 4$:-

$A = 16.8$ $\quad B = 1.067$ $\quad C = 5120.07$ $\quad D = 66561.067$

$S(1) = 7.251$

$S(2) = 996.502$

$S(3) = 7787.754$

$S(4) = 71699.004$

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Distribution seems polynomial, rest of graph dominating as (T^3) increases with t_{yr}.

